HKUST MATH 2431 problem set 2

3.

3. (*) **Rolling dice a random number of times** [4 Points]

A fair die is rolled N times, where N is a random number with values in $\mathbb{N} = \{1, 2, ...\}$. We call A_n the event that $N = n$ and S_M the event that the sum of all numbers shown is equal to $M \in \mathbb{N}$. We will assume the following:

(1) For every die roll, every number shows up with the same probability (the die is fair),

(2) For $n \in \mathbb{N}$, $\mathbf{P}[A_n] = \frac{1}{2^n}$.

(a) Determine an appropriate sample space Ω and a map

$$\mathbf{P} : \mathscr{P}(\Omega) \to [0, 1],$$

such that $(\Omega, \mathscr{P}(\Omega), \mathbf{P})$ is a probability space and (1) and (2) are fulfilled.

3(a) $\Omega = \{(x_1, \dots, x_n) : n \in \mathbb{N}_{\neq 0},\ x_i \in \{1, \dots, 6\}\}$

$\mathcal{P}(\Omega) = 2^{\Omega}$ (Ω is countable, can take its power set)

$P(\{w\}) = \frac{1}{2^{|w|}} \cdot \frac{1}{6^{|w|}} = \frac{1}{12^{|w|}}$

where $|w|$ is the card of w

Calculate the probabilities

(b) $\mathbf{P}[A_2 | S_4]$,

3(b) Partitions of 4

$$
\begin{aligned}
4 &= 1+1+1+1 \rightarrow 1 \text{ outcome, each of prob } 12^{-4} \\
&= 1+1+2 \rightarrow 3 \text{ outcome, each of prob } 12^{-3} \\
&= 1+3 \rightarrow 2 \text{ outcome, each of prob } 12^{-2} \\
&= 2+2 \rightarrow 1 \text{ outcome, each of prob } 12^{-2} \\
&= 4 \rightarrow 1 \text{ outcome, each of prob } 12^{-1}
\end{aligned}
$$

(c) $\mathbf{P}[S_4|A_e]$, where A_e is the event that N is an even number.

Thus $\mathbb{P}[A_2 | S_4]$

$$= \frac{12^{-2}(3)}{12^{-4}(1) + 12^{-3}(3) + 12^{-2}(3) + 12^{-1}(1)}$$

$$= \frac{12^{-4}(432)}{12^{-4}(1+36+432+1728)}$$

$$= \frac{432}{2197} \approx 0.196721$$

(c) $\mathbf{P}[S_4|A_e]$, where A_e is the event that N is an even number.

3(c) Reusing the partitions of 4 in 3(b), we have

$$\mathbb{P}[A_e] = \sum_{k=1}^{\infty} \frac{1}{2^{2k}} = \sum_{k=1}^{\infty} \frac{1}{4^k} = \frac{(1/4)}{(1-1/4)} = \frac{1/4}{3/4} = \frac{1}{3}$$

$$\mathbb{P}[S_4 \mid A_e] = \frac{12^{(-2)}(3) + 12^{(-4)}(1)}{1/3}$$

$$= (3)\,12^{(-4)}(1+432)$$

$$= \frac{433}{6912} \approx 0.0626447$$

4.

4. (∗) **Sample size vs. signal strength** [4 Points]

An urn contains 3 red and 3 blue balls. One of these balls, called **A**, is selected uniformly at random and permanently removed from the urn without the color of this ball being shown to an observer. This observer may now draw successively – uniformly at random and with replacement – a number of individual balls (one at a time) from among the five remaining balls.

Peter may draw a ball from the urn six times, and each time the ball he draws turns out to be red. Paula may draw a ball from the urn 600 times; 303 times she draws turns out to be red, and 297 times she draws a blue ball.

Let R be the event that **A** is red, P_1 be the event that Peter's sequence is realized, and P_2 be Paula's sequence is realized.

(a) Calculate $\mathbf{P}[R]$, $\mathbf{P}[R|P_1]$, and $\mathbf{P}[R|P_2]$.

4 (a) $\mathbb{P}[R] = \frac{3}{6} = \frac{1}{2}$ (same for $\mathbb{P}[B]$)

$$\mathbb{P}[R \mid P_1] = \frac{\mathbb{P}[P_1 \mid R]\,\mathbb{P}[R]}{\underbrace{\mathbb{P}[P_1]}_{=\mathbb{P}[P_1 \mid R]\mathbb{P}[R] + \mathbb{P}[P_1 \mid B]\mathbb{P}[B]}} = ?$$

$\mathbb{P}[P_1 \mid R] = \left(\frac{2}{5}\right)^6$ $\qquad$ $\mathbb{P}[P_1 \mid B] = \left(\frac{3}{5}\right)^6$

$$\mathbb{P}[R \mid P_1] = \frac{\left(\frac{2}{5}\right)^6\left(\frac{1}{2}\right)}{\left(\frac{2}{5}\right)^6\left(\frac{1}{2}\right) + \left(\frac{3}{5}\right)^6\left(\frac{1}{2}\right)}.$$

$$= \frac{2^6}{2^6 + 3^6} = \frac{64}{793}$$

$$\mathbb{P}[R \mid P_2] = \frac{\mathbb{P}[P_2 \mid R]\,\mathbb{P}[R]}{\underbrace{\mathbb{P}[P_2]}_{=\mathbb{P}[P_2 \mid R]\mathbb{P}[R] + \mathbb{P}[P_2 \mid B]\mathbb{P}[B]}} = ?$$

$\mathbb{P}[P_2 \mid R] = \binom{600}{303}\left(\frac{2}{5}\right)^{303}\left(\frac{3}{5}\right)^{297}$ $\qquad$ $\mathbb{P}[P_2 \mid B] = \binom{600}{303}\left(\frac{3}{5}\right)^{303}\left(\frac{2}{5}\right)^{297}$

$$\mathbb{P}[R \mid P_2] = \frac{\binom{600}{303}\left(\frac{2}{5}\right)^{303}\left(\frac{3}{5}\right)^{297}}{\binom{600}{303}\left(\frac{2}{5}\right)^{303}\left(\frac{3}{5}\right)^{297} + \binom{600}{303}\left(\frac{3}{5}\right)^{303}\left(\frac{2}{5}\right)^{297}}$$

$$= \frac{2^{303}3^{297}}{2^{303}3^{297} + 3^{303}2^{297}}$$

$$= \frac{2^6}{2^6 + 3^6}$$

$$= \frac{64}{793}$$

(b) Interpret your result.

4(b) Interestingly, $P[R|P_1] = P[R|P_2] = \frac{64}{793}$

Only the absolute difference in the number of red and blue balls matter.

We can interpret it like so:

- calculating $P[R|P_1]$ and $P[R|P_2]$ can be interpreted in statistics as by sampling from the urn, to try to infer which ball is drawn,
- Notice the drawing order does not matter, Drawing a red or blue ball at the 1st or 100th draw both tells the exact same information about the urn, because the draws are mutually independent. As a result, only the number of red and blue draws is sufficient to capture all information about the urn from the draws.
- Finally, drawing a red ball increases the confidence that A is blue by as much as drawing a blue ball increases the confidence that A is red.

This can be explained by symmetry of the setup (the setup is the same if you swap all instances of "red" and "blue" in the question). This can also be considered intuitively: considering drawing one red and one blue. Then this draw tells you no information about whether A = red or A = blue, because in both prior situations, the likelihood is the same, so the posterior is also the same. As a result, we can say drawing a red ball "cancels out" drawing a blue ball. So only the differences in the # of red and blue balls matters (has valuable information about the urn).